DynÉire: A Dynamic Overlapping Generations Model for Irish Fiscal Policy Analysis

Giorgia Conte[*] Joseph Kopecky[*] Barra Roantree[*]

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Abstract

This paper presents DynÉire, a dynamic overlapping generations (OG) general equilibrium model calibrated to the Irish economy. Building on the OG-Core computational framework developed by the Policy Simulation Library, we construct a dynamic, heterogeneous agent, general equilibrium model to study Ireland's unique fiscal characteristics informing macroeconomic general equilibrium adjustments with micro-founded tax responses. To do this, the we integrate microsimulation evidence from EUROMOD to estimate age- and income-varying tax functions that capture the full progressivity of the Irish tax system. We present results from three policy experiments: (1) a 3 percentage point reduction in employer payroll taxes; (2) complete removal of the Universal Social Charge; and (3) a sudden loss of multinational corporate (MNC) tax revenue. This model allows us to calculate macroeconomic effects of policy change that account for endogenous response of households to changes in economics conditions while also providing a framework for welfare considerations. In our three illustrative policy experiments we show that USC abolition generates the large welfare gains (+2.27% consumption equivalent variation), with output increasing 2.63% through higher labor supply and capital accumulation. Employer PRSI reduction produces similar gains (+1.89% CEV). Our corporate tax exercise demonstrates the usefulness of the model to provide a laboratory to study policy options in response to exogenous revenue loss, capturing costly fiscal adjustments that would be needed to close fiscal gaps.

[*]Department of Economics, Trinity College Dublin

1. Introduction

Ireland faces distinctive fiscal policy challenges stemming from its position as a small open economy with a unique relationship with foreign multinational corporations. Tax policy decisions have complex general equilibrium effects that static analysis cannot fully capture. This paper introduces DynÉire, an overlapping generations model calibrated to the Irish economy, designed to analyze the dynamic macroeconomic and distributional consequences of fiscal policy reforms.

We propose a flexible approach to policy simulation in the Irish context, building on work of existing models both within the Irish context and abroad. In particular, we construct a dynamic general equilibrium model with overlapping-generations of heterogeneous households. This primarily extends work conducted for the United States by DeBacker et al. (2019), specify a model that can ingest rich micro-simulated tax policy data to capture detailed income heterogeneity and real-world demographic movements in the context of a general equilibrium where agents endogenously react to both changes in policies as well as long term movements in underlying population. We calibrate and extend the OG-USA model of Evans and DeBacker (2018), to the Irish context. This model allows finitely lived households to optimally choose consumption, labor supply, and savings and to study the short term implications of a policy change in addition to the long term adjustment towards new equilibria. Key adaptations include: calibrating to Irish fertility, mortality, and migration patterns; adjusting tax structures to reflect the income tax, USC, and PSRI systems in Ireland; and enhancing the model's ability to deal with the outsized role of corporate taxes to more explicitly understand key vulnerabilities.

Such heterogeneous agent policy models have become and important tool in macroeconomics, with many agencies building heterogeneous agent new Keynesian (HANK) models[1] to study policy. There are two important reasons to believe that heterogeneity is important. First, existing work such as Broer et al. (2023) suggests potentially large differences in policy impact in the presence of such heterogeneity relative to representative agent (and two-agent) counterparts. Secondly, distributional impacts of policy are an important area of study in themselves, with heterogeneous agent models allowing researchers to understand model implied behavioral changes in the face of policy. To our knowledge the only existing general equilibrium model for Ireland that incorporates heterogeneity is the small open-economy (SOE) model of Varthalitis (2019) which allows for two types of agents. Our model allows for similar SOE structure, while bringing significantly more heterogeneity into the household problem, which in turn allows for the study of tax/benefit

[1]For monetary policy see for example, Kaplan et al. (2018) and for fiscal see, Broer et al. (2023)

changes in a granular, and more realistic way.

We use micro simulation data from EUROMOD[2] to fit tax functions that vary across both age and income dimensions, which are then used in the GE model. A key contribution relative to existing approaches within Ireland is the ability to use these micro-founded simulations of the underlying tax/benefit environment which are then passed to a model that retains a large portion of the heterogeneity. Policy changes are simulated in the micro-model before being passed to the general equilibrium model where transition paths between two policy environments can be calculated.

The strength of this approach will be in understanding potential behavioral implications across the age-income distribution in response to tax shocks and their short term implications, as well as medium-to-long run implications of policy on macroeconomic aggregates and welfare. We envision this as complementary to existing models such as the COSMO model of Bergin et al. (2017) and existing GE models such as Clancy et al. (2016) and Varthalitis (2019). Moreover, we plan to establish this as an open source project in similar spirit to the OG-USA project of Evans and DeBacker (2018) with scope for a wide range of applications and extensions to suit the needs of interested researchers. Because the Irish case is unique along a number of dimensions we hope this will serve not only as a useful tool for policymakers, but also an important ground for economic research more broadly.

Ultimately the goal of this project is twofold. The first is to produce research for which the Irish context serves as a useful test case for deeper economic questions. The second is to be able to set up an ongoing framework to participate in the policy discussion in Ireland as a complement to the existing fiscal models. In this preliminary work we present results from three policy experiments that serve to showcase the features of this model and the types of contributions that it is capable of making to ongoing policy debates.

1. **Employer PRSI reduction**: Simulating cuts to the 11.05% employer contribution
2. **USC abolition**: Evaluating the removal of the Universal Social Charge
3. **MNC revenue shock**: Analyzing the consequences of a sudden loss of multinational corporate tax revenue and comparing fiscal adjustment mechanisms (VAT increases, payroll tax increases, spending cuts, transfer cuts, and mixed adjustments)

The remainder of this paper is organized as follows. Section 2 presents the theoretical framework. Section 3 describes the Irish calibration. Sections 4–6 present the policy experiments. Section 8 concludes.

[2] Sutherland and Figari (2013) and Sutherland et al. (2018)

2. Model Structure

The DynÉire model is a dynamic overlapping generations general equilibrium model in the tradition of Auerbach and Kotlikoff (1987). This section presents the key model components. It builds on the life-cycle overlapping generations framework of the open source OG-USA model of DeBacker et al. (2019). Ultimately its strength is to incorporate micro-foundations, through carefully calibrated heterogeneous agents, into the framework of a general equilibrium model that allows for endogenous response of households to policy shifts.

2.1. Demographics

The economy is populated by S generations of agents who live with certainty from age $E+1$ (model age 1) to age $E+S$. At each age s, agents face mortality risk ρ_s. The population evolves according to:

$$\omega_{t+1,s+1} = (1-\rho_s)\omega_{t,s} + i_{t+1,s+1} \tag{1}$$

where $\omega_{t,s}$ is the measure of age-s agents at time t and $i_{t,s}$ represents immigration. For the youngest cohort:

$$\omega_{t+1,1} = (1-\rho_0)\sum_{s=1}^{S} f_s\omega_{t,s} + i_{t+1,1} \tag{2}$$

where f_s is the age-specific fertility rate.

In steady state, the population growth rate is:

$$1 + g_n = \frac{\sum_{s=1}^{S}\omega_{SS,s}}{\sum_{s=1}^{S}\omega_{SS,s-1}(1-\rho_{s-1})} \tag{3}$$

2.2. Households

Households are heterogeneous in two dimensions: age $s \in \{1,2,...,S\}$ and ability type $j \in \{1,2,...,J\}$. The measure of type-j agents in the population is λ_j, with $\sum_j \lambda_j = 1$.

2.2.1 Preferences

Households maximize expected lifetime utility:

$$U = \sum_{s=1}^{S} \beta^{s-1} \left(\prod_{u=1}^{s-1} (1-\rho_u) \right) u(c_{s,j,t+s-1}, n_{s,j,t+s-1}) \tag{4}$$

where $c_{s,j,t}$ is consumption and $n_{s,j,t}$ is labor supply. The period utility function is:

$$u(c,n) = \frac{c^{1-\sigma}}{1-\sigma} - \chi_n^s \frac{n^{1+\frac{1}{\gamma}}}{1+\frac{1}{\gamma}} \tag{5}$$

where σ is the coefficient of relative risk aversion, γ is the Frisch elasticity of labor supply, and χ_n^s is the age-specific disutility of labor.

2.2.2 Budget Constraint

The household's flow budget constraint is:

$$(1+\tau_c)c_{s,j,t} + b_{s+1,j,t+1} = (1-\tau_{s,j,t}^{ETR})y_{s,j,t} + (1+r_t)b_{s,j,t} + T_{s,j,t} + BQ_{s,j,t} \tag{6}$$

where, τ_c is the consumption tax rate, $b_{s,j,t}$ is beginning-of-period savings, $\tau_{s,j,t}^{ETR}$ is the effective tax rate on total income, $y_{s,j,t} = w_t e_{s,j} n_{s,j,t} + r_t b_{s,j,t}$ is total income, $e_{s,j}$ is the age- and ability-specific earnings profile, $T_{s,j,t}$ are government transfers, and $BQ_{s,j,t}$ are bequests received.

2.2.3 Optimality Conditions

The household's Euler equation for saving is:

$$u_c(c_{s,j,t}, n_{s,j,t}) = \beta(1-\rho_s)(1 + r_{t+1}(1-\tau_{t+1}^{MTR,y}))u_c(c_{s+1,j,t+1}, n_{s+1,j,t+1}) \tag{7}$$

The intratemporal labor supply condition is:

$$\chi_n^s n_{s,j,t}^{1/\gamma} = \frac{(1-\tau_{s,j,t}^{MTR,x})w_t e_{s,j}}{(1+\tau_c)} u_c(c_{s,j,t}, n_{s,j,t}) \tag{8}$$

where $\tau^{MTR,x}$ and $\tau^{MTR,y}$ are marginal tax rates on labor and capital income, respectively.

2.3. Firms

A representative firm produces output using a Cobb-Douglas production function:

$$Y_t = Z_t K_t^{\gamma} L_t^{1-\gamma} \tag{9}$$

where Z_t is total factor productivity growing at rate g_y, K_t is the capital stock, and L_t is effective labor.

Profit maximization yields factor demands:

$$r_t + \delta = \gamma \frac{Y_t}{K_t}(1 - \tau_b) \tag{10}$$

$$w_t = (1 - \gamma)\frac{Y_t}{L_t}(1 - \tau_{\text{payroll}}) \tag{11}$$

where δ is the depreciation rate, τ_b is the corporate tax rate, and τ_{payroll} is the employer payroll tax rate.

2.4. Government

The government collects taxes, provides transfers, purchases goods, and issues debt. The government budget constraint is:

$$G_t + T_t + (1 + r_t^{gov})D_t = R_t + D_{t+1} \tag{12}$$

where G_t is government consumption, T_t is total transfers, D_t is government debt, and R_t is total tax revenue:

$$R_t = R_t^{income} + R_t^{payroll} + R_t^{consumption} + R_t^{corporate} + R_t^{bequest} \tag{13}$$

Government spending and transfers are assumed to be fixed shares of output:

$$G_t = \alpha_G Y_t \tag{14}$$

$$T_t = \alpha_T Y_t \tag{15}$$

2.5. Market Clearing

The goods market clears:

$$Y_t = C_t + I_t + G_t \tag{16}$$

The capital market clears:

$$K_{t+1} = B_{t+1} + D^f_{t+1} \tag{17}$$

where B_{t+1} is aggregate domestic savings and D^f_{t+1} is net foreign capital inflows.

The labor market clears:

$$L_t = \sum_{s=1}^{S} \sum_{j=1}^{J} \omega_{t,s} \lambda_j e_{s,j} n_{s,j,t} \tag{18}$$

2.6. Equilibrium

A competitive equilibrium consists of sequences of household allocations $\{c_{s,j,t}, n_{s,j,t}, b_{s,j,t}\}$, firm decisions $\{K_t, L_t\}$, prices $\{r_t, w_t\}$, and government policies $\{G_t, T_t, D_t, \tau_t\}$ such that:

1. Households maximize utility subject to budget constraints
2. Firms maximize profits
3. The government satisfies its budget constraint
4. All markets clear

3. Calibration to Ireland

This section describes the calibration of DynÉire to the Irish economy. Table 4 summarizes the key parameters.

3.1. Model Dimensions

The model is calibrated with $S = 80$ life periods corresponding to ages 20–100, $J = 7$ ability types representing different positions in the income distribution, and $T = 320$ time periods for transition dynamics. The baseline year is 2024. The ability type distribution (λ_j) follows:

Group	Bottom 25%	25–50%	50–70%	70–80%	80–90%	90–99%	Top 1%
λ_j	0.25	0.25	0.20	0.10	0.10	0.09	0.01

3.2. A note on GDP vs. GNI* in the Irish Context

A key feature of the Irish calibration is adjusting for the fact that Ireland's GDP is significantly distorted by multinational corporate activities. We calibrate fiscal ratios using Modified Gross National Income (GNI*), which excludes a number of factors, but critically retained earnings of re-domiciled companies. In 2024, Irish GDP was approximately €550 billion, while GNI* was approximately €330 billion—a ratio of about 0.60. Table 1 shows how this affects the calibration of key fiscal ratios.

Table 1: *GDP vs. GNI* Fiscal Ratio Comparison*

Ratio	GDP basis	GNI* basis
Government debt-to-output	44%	73%
Government consumption share	12%	20%
Transfer share	10%	17%

3.3. Demographics

Demographic parameters are calibrated using data from the Central Statistics Office (CSO) and the Human Mortality Database. Age-specific fertility rates are taken from CSO Vital Statistics (series VSA02). Ireland's total fertility rate was approximately 1.7 in 2024, with peak fertility occurring at ages 32–33—later than the United States (age 28). Age-specific mortality rates are constructed from CSO life tables and Human Mortality Database data. Irish life expectancy at birth was approximately 82 years in 2024, with infant mortality of 0.30%. Net migration rates are calibrated as a residual from the population accounting identity, using CSO Population Estimates (series PEA11 and PEA15). Ireland experienced significant net immigration, particularly for ages 20–35. The calibrated steady-state population growth rate is $g_n = 0.8\%$ per annum.

3.4. Macroeconomic Parameters

Key macroeconomic parameters are summarized in Table 2.

3.5. Earnings Profiles

Age-specific earnings profiles $e_{s,j}$ are estimated using CSO Earnings and Labour Costs data and EU-SILC microdata. The functional form follows:

$$\ln(e_{s,j}) = \alpha_j + \beta_1 s + \beta_2 s^2 + \beta_3 s^3 \tag{19}$$

where s is age. Separate profiles are estimated for each ability type j. Earnings peak at approximately age 50–55, with higher ability types exhibiting steeper earnings growth early in life. Earnings decline smoothly after the retirement age of 66.

Table 2: *Macroeconomic Parameters*

Parameter	Description	Value	Source
γ	Capital share of income	0.35	CSO National Accounts
δ	Annual depreciation rate	0.05	Standard calibration
g_y	Annual productivity growth	0.015	GNI*-basis trend
σ	Risk aversion	1.5	Literature
γ^{Frisch}	Frisch elasticity	0.40	Chetty et al. (2012)
β	Annual discount factor	0.96	Calibrated
Government Parameters (GNI basis)*			
α_G	Government consumption/output	0.12	CSO Government Accounts
α_T	Transfers/output	0.10	DSP statistics
D_0/Y_0	Initial debt ratio	0.44	NTMA
Tax Rates (2024)			
τ_c	VAT rate	0.23	Revenue Commissioners
τ_b	Corporate tax rate	0.125	Revenue Commissioners
$\tau_{payroll}^{employer}$	Employer PRSI	0.1105	DSP
$\tau_{payroll}^{employee}$	Employee PRSI	0.04	DSP
τ_{bq}	Capital Acquisitions Tax	0.33	Revenue Commissioners

3.6. Tax System

The Irish tax system is approximated using effective and marginal tax rates. Table 3 summarizes the full progressive structure as of 2024. Income tax applies at a standard rate of 20% on the first €42,000 (for single filers), with a higher rate of 40% above the standard rate band. The Universal Social Charge (USC) is levied at progressive rates ranging from 0.5% to 8%. Social insurance contributions (PRSI) are paid by both employees (4%) and employers (11.05% standard rate).

3.7. Transfer and Bequest Distributions

Government transfers are distributed across age-ability groups according to the matrix $\eta_{s,j}$, calibrated to reflect higher transfers to lower-income groups, elevated transfers to young adults through unemployment benefits, and universal child benefit affecting middle-age groups. Bequests are distributed according to $\zeta_{s,j}$, with inheritance receipt concentrated at ages 45–70 and skewed toward higher-wealth groups, reflecting intergenerational wealth persistence.

Table 3: *Irish Tax Structure (2024)*

Tax	Band/Description	Rate
Income Tax		
	First €42,000 (single)	20%
	Above €42,000	40%
Universal Social Charge (USC)		
	First €12,012	0.5%
	€12,012 – €28,700	2%
	€28,700 – €70,044	3%
	Above €70,044	8%
PRSI (Social Insurance)		
	Employee contribution	4%
	Employer contribution	11.05%
Baseline Effective Rates		
	Average ETR	25%
	Average MTR on labor (MTR_x)	40%
	MTR on capital income (MTR_y)	33%

Table 4: *Summary of Calibrated Parameters*

Parameter	Symbol	Value	Source
Life periods	S	80	Ages 20–100
Ability types	J	7	Income distribution
Time periods	T	320	Transition path
Starting age		20	
Retirement age		66	State pension age
Frisch elasticity	γ^{Frisch}	0.40	Literature
Risk aversion	σ	1.5	Literature
Annual depreciation	δ	0.05	Standard
Capital share	γ	0.35	CSO
TFP growth (GNI*)	g_y	1.5%	Trend
Population growth	g_n	0.8%	CSO demographics
Mean income		€60,000	CSO

3.8. Tax Function Estimation

A key feature of DynÉire is the use of estimated tax functions that capture the full progressivity and variation in effective and marginal tax rates across age and income groups. Rather than imposing simple average tax rates, we estimate flexible tax functions from microsimulation data that map total income to tax rates for each age cohort.

The methodology follows the approach of DeBacker et al. (2019). We use EUROMOD, a tax-benefit microsimulation model maintained by the European Commission, to simulate the Irish tax system for a representative sample of households. For each household, EUROMOD calculates total taxes paid given their income, yielding effective tax rates (ETR) and marginal tax rates (MTR) on both labor and capital income.

We then estimate Devereux-Griffith-Pearce (DEP) tax functions that provide a smooth approximation to the discrete microsimulation output:

$$\tau(y) = D + \max\{0, Ay + B\}^{Cy+F} \tag{20}$$

where y is total income and the parameters $\{A, B, C, D, F\}$ are estimated separately for each age group. This functional form flexibly captures the progressive structure of the Irish tax system, including the interaction of income tax bands, USC thresholds, and PRSI contributions.

Figure 1 illustrates the estimated tax functions for age 43, comparing the baseline tax system (with USC) to a reform scenario (without USC). The top panels show effective tax rates, while the middle panels show marginal tax rates on labor income. The scatter points represent individual observations from the microsimulation, and the red lines show the fitted DEP functions. The bottom panels compare the baseline and reform functions directly, highlighting the effect of USC removal on both ETR and MTR across the income distribution.

The estimated tax functions are then integrated into the general equilibrium, where they determine households' budget constraints and through this affect household: labor supply, saving, and consumption decisions. In this way we can not only micro-found the model through agent heterogeneity of skill (which here maps to income), but to also understand at a granular level how a macro change in tax policy affects the effective and marginal rates faced across the entire age-income distribution. This approach combines the institutional detail of microsimulation with the behavioral responses of general equilibrium modeling.

In what follows we estimate three policy experiments. These are meant to demonstrate the flexibility of the modeling and are, at this stage, preliminary. However, we think they provide a great deal of insight into important questions facing policymakers.

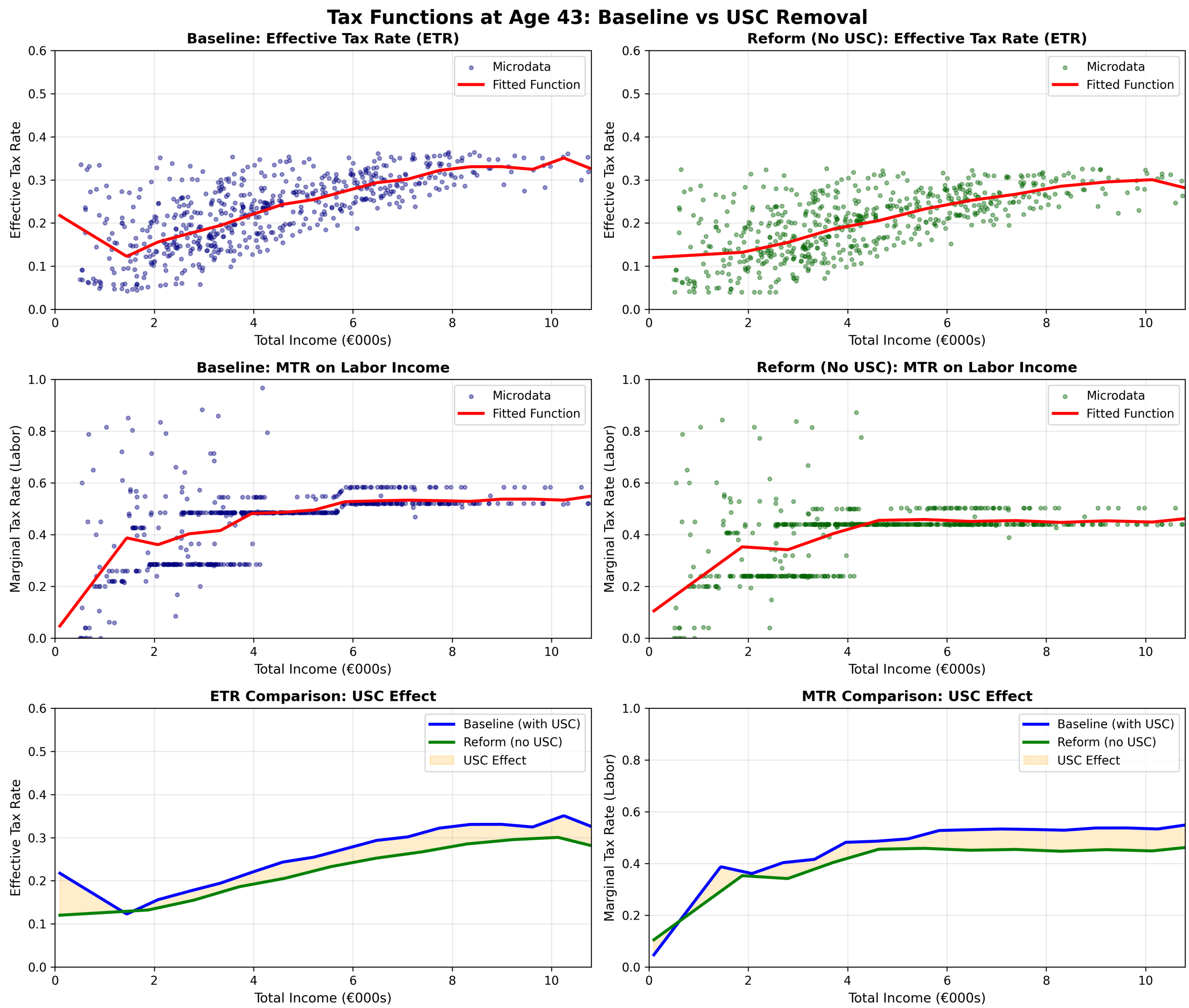

Figure 1: *Tax Functions at Age 43: Baseline vs. USC Removal. Note: The figure shows effective tax rates (top row) and marginal tax rates on labor income (middle row) for the baseline system (left) and reform without USC (right). Scatter points are microsimulation observations; red lines are fitted DEP functions. Bottom panels compare baseline and reform functions directly, with the shaded area representing the USC effect.*

4. Policy Experiment 1: Employer PRSI Reduction

Employer social insurance contributions (PRSI) represent a significant labor tax wedge in Ireland. This experiment examines the effects of reducing employer PRSI. This is a logical first step in understanding model dynamics as it works through firms and does not (at least in the present calibration) require a shift in age-income tax burden simulated through a Euromod.[3]

4.1. Policy Context

Current employer PRSI structure (2024) can be described by: a standard rate: 11.05% on all earnings, lower rate: 8.8% for employees earning under €441/week, no upper earnings limit, and total employer PRSI revenue: approximately €12–13 billion.

Employee PRSI is 4%, for a combined payroll tax rate of approximately 15%.

4.2. Modeling Approach

Employer PRSI affects the model through the firm's labor demand, given by:

$$w = (1-\gamma)\frac{Y}{L}(1-\tau_{payroll}) \tag{21}$$

A reduction in $\tau_{payroll}$ reduces the wedge between the marginal product of labor and the wage received by workers. This affects labor demand (firms face lower costs), labor supply (if wage incidence is on workers), capital-labor ratio, as well as output and consumption/

4.3. Policy Experiment

We consider a 3 percentage point reduction in employer PRSI:

- Baseline rate: 11.05%
- Reform rate: 8.05%
- Sensitivity analysis: 1pp, 2pp, 3pp, 5pp cuts

The model operates under budget balance, so reduced payroll tax revenue is offset by adjustments to other fiscal parameters.

[3] This of course may need to be adjusted going forward if the incidence of this tax does have a knock-on affect that we should model here, but at the moment we're treating it as a pure firm-side shock.

4.4. Results

Table 5 presents the steady-state macroeconomic effects of the 3 percentage point employer PRSI reduction. The reform leads to a 1.95% increase in output, driven by both higher labor supply (+1.89%) and capital accumulation (+2.07%). The reduction in the labor tax wedge stimulates employment, while the resulting higher wages encourage saving and investment.

Table 5: *Macroeconomic Effects: Payroll Cut*

Variable	Baseline	Reform	% Change
Output (Y)	0.6307	0.6430	+1.95%
Consumption (C)	0.3861	0.3934	+1.89%
Capital (K)	2.2584	2.3052	+2.07%
Labor (L)	0.3173	0.3233	+1.89%
Interest Rate (r)	0.0452	0.0451	-0.24%
Wage (w)	1.2919	1.2927	+0.06%

Table 6 summarizes the welfare effects. The consumption equivalent variation (CEV) of +1.89% indicates a substantial welfare gain from the policy.

Table 6: *Scenario Comparison: Payroll*

Scenario	ΔY (%)	ΔC (%)	CEV (%)
payroll_cut	+1.95%	+1.89%	+1.89%

Note: All values relative to baseline. CEV = Consumption Equivalent Variation.

5. Policy Experiment 2: USC Abolition

The Universal Social Charge (USC) was introduced in 2011 as an emergency measure during the financial crisis. Debate continues about whether it should be abolished. Since this directly affects household income tax burdens we model this through the microsimulations, which can be seen in Figure 1, but these must also feed into our general equilibrium framework. DynÉire can speak directly to the long term effects of such a cut.

5.1. Policy Context

Current USC structure (2024):

Income Band	Rate
First €12,012	0.5%
€12,012 – €28,700	2%
€28,700 – €70,044	3%
Above €70,044	8%

Key facts:

- USC revenue: approximately €4–5 billion annually
- This represents about 12–15% of income tax revenue
- The USC is progressive, with higher rates on higher incomes
- Unlike PRSI, USC applies to all income (including self-employment and investment income)

5.2. Modeling Approach

We model USC abolition using tax functions estimated from EUROMOD microsimulation data. This approach captures the full progressivity and variation in tax rates across age and income groups, rather than using simple average rates.

The methodology follows three steps:

1. **EUROMOD simulation**: We simulate baseline (with USC) and reform (without USC) scenarios using EUROMOD, Ireland's official tax-benefit microsimulation model. This generates effective tax rates (ETR) and marginal tax rates (MTR) for a representative sample of Irish households. See Figure 1.
2. **Tax function estimation**: We estimate Devereux-Griffith-Pearce (DEP) tax functions that map income to tax rates, allowing rates to vary by age and income level. The DEP functional form provides a flexible approximation to the progressive tax schedule. See Figure 1.
3. **General equilibrium integration**: The estimated tax functions are incorporated into the DynÉire model, where they determine households' budget constraints and affect labor supply, saving, and consumption decisions.

What we can learn from these microsimulations is that USC abolition affects the model through effective tax rates (average ETR falling from 21.6% to 19.2%, 2.4 percentage point reduction), as well as marginal tax rates (average MTR on labor falls from 40.6% to 36.4%, a

4.2 percentage point reduction). The general equilibrium modelling allows us to further understand labor supply incentives and response (lower MTR increases labor supply through substitution effects), as well as the impact of saving incentives (lower ETR increases disposable income, increasing saving and capital accumulation).

This integrated microsimulation-GE approach combines the institutional detail of EUROMOD with the behavioral and general equilibrium structure of DynÉire, providing more accurate estimates than either model alone.

5.3. Policy Experiment

Complete removal of the USC. We estimate effective and marginal tax rates using EUROMOD microsimulation data for baseline (with USC) and reform (without USC) scenarios, then fit age- and income-varying tax functions using the Devereux-Griffith-Pearce (DEP) functional form:

Parameter	With USC	Without USC
Average ETR	21.6%	19.2%
Average MTR (labor)	40.6%	36.4%
Average MTR (capital)	33%	33%

The EUROMOD-estimated tax functions show that USC removal reduces the effective tax rate by 2.4 percentage points and the marginal tax rate on labor income by 4.2 percentage points. These changes vary across age and income groups, with higher-income earners experiencing larger reductions due to the progressive structure of the USC.

5.4. Predicted Effects

Before presenting the full model results, we can use standard formulas to predict the direction and approximate magnitude of effects. This is useful to sanity check our later results.

Labor supply response:

$$\Delta L/L \approx -\gamma^{Frisch} \times \frac{\Delta\tau^{MTR}}{1-\tau^{MTR}} = -0.4 \times \frac{-0.04}{0.60} \approx +2.7\% \tag{22}$$

Saving response:

$$\Delta K/K \approx \text{IES} \times \frac{-\Delta\tau^{ETR}}{1-\tau^{ETR}} = 0.5 \times \frac{0.04}{0.75} \approx +2.7\% \tag{23}$$

Output response (Cobb-Douglas):

$$\Delta Y/Y \approx \gamma \cdot \frac{\Delta K}{K} + (1-\gamma) \cdot \frac{\Delta L}{L} \approx +2.5\% \tag{24}$$

These simple calculations suggest USC removal should generate output gains of approximately 2–3%. The full computational model presented in the next section solves the complete system of household optimality conditions, firm problems, government budget constraints, and market clearing conditions to obtain precise equilibrium outcomes accounting for agent heterogeneity, progressive tax structures, and all general equilibrium feedback effects.

5.5. Results

Table 7 presents the steady-state macroeconomic effects of USC abolition. The reform generates substantial positive effects on output, capital, and labor supply, driven by reduced tax distortions on both labor supply and saving decisions.

Table 7: *USC Abolition: Steady-State Effects*

Variable	Baseline	Reform	% Change
Output (Y)	0.684	0.702	+2.63%
Consumption (C)	0.463	0.474	+2.27%
Capital Stock (K)	2.312	2.395	+3.59%
Labor Supply (L)	0.355	0.362	+2.12%
Interest Rate (r)	0.0414	0.0408	-1.52%
Wage Rate (w)	1.329	1.344	+1.14%
Tax Revenue	0.130	0.118	-8.90%
Government Debt (D)	0.301	0.309	+2.64%
CEV (Welfare)	—	—	+2.27%

Note: Results based on age-specific tax functions estimated from EUROMOD microsimulation data. CEV is the consumption equivalent variation measuring welfare change.

GDP increases by 2.63% in the long run, driven by both increased labor supply (+2.12%) and capital accumulation (+3.59%). The output effect is close to the analytical estimate of +2.5% derived above. The 4.2 percentage point reduction in marginal tax rates on labor generates a 2.12% increase in labor supply, implying an effective Frisch elasticity consistent with the calibrated value of 0.40. The reduction in effective tax rates increases disposable income and saving, leading to 3.59% higher capital stock in steady state, which crowds in investment and raises the capital-labor ratio.

Tax revenue falls by 8.90%, reflecting both the direct revenue loss from USC removal and partially offsetting gains from the expanded tax base. The fiscal cost is less than the mechanical revenue loss due to positive behavioral responses. The consumption equivalent variation (CEV) is +2.27%, indicating substantial welfare gains as households benefit from higher disposable income, reduced tax distortions, and increased consumption. The equilibrium interest rate falls slightly (-1.52%) as increased capital accumulation reduces the marginal product of capital.

6. Policy Experiment 3: MNC Revenue Withdrawal

Ireland's exceptional dependence on multinational corporate (MNC) tax revenue represents a significant fiscal vulnerability. This section examines the macroeconomic consequences of a sudden loss of MNC tax revenue and compares alternative fiscal adjustment mechanisms.

6.1. Policy Context

Key features of Ireland's corporate tax revenue:

- Total corporate tax revenue: approximately €20–25 billion (2024)
- This represents roughly 20% of total tax revenue—unusually high by international standards
- Approximately 75% of corporate tax revenue comes from multinational corporations
- MNC tax revenue is highly concentrated: the top 10 companies account for over 50% of corporate tax receipts
- This revenue is vulnerable to changes in international tax rules, MNC location decisions, and global economic conditions

The OECD/G20 global minimum tax (Pillar Two) and potential changes to transfer pricing rules pose risks to Ireland's MNC tax base. Additionally, individual company decisions to relocate or restructure could lead to sudden revenue losses.

6.2. Modeling Approach

We model an exogenous, permanent loss of MNC corporate tax revenue. This captures scenarios where:

- Major MNCs relocate operations or profits to other jurisdictions
- International tax reforms reduce the attractiveness of booking profits in Ireland
- A small number of large taxpayers experience significant business disruptions

The shock is modeled as an exogenous reduction in government revenue equal to 3.75% of GDP (approximately 75% of corporate tax revenue, or €12–15 billion in 2024 terms). Budget balance is restored through one of five fiscal adjustment mechanisms:

1. **VAT increase**: Raising the consumption tax rate
2. **Payroll tax increase**: Raising employer PRSI contributions
3. **Spending cuts**: Reducing government consumption
4. **Transfer cuts**: Reducing social transfers
5. **Mixed adjustment**: A combination of VAT, spending, and transfer adjustments (one-third each)

6.3. Policy Experiment

We simulate a permanent loss of MNC revenue equal to 3.75% of GDP. The required adjustments under each mechanism are:

Adjustment	Instrument	Change
VAT	Consumption tax rate	+7.5 pp (23% → 30.5%)
Payroll	Employer PRSI	+6.25 pp (11% → 17.3%)
Spending	Govt consumption/GDP	−3.75 pp
Transfers	Transfers/GDP	−3.75 pp
Mixed	VAT, spending, transfers	+2.5 pp, −1.25 pp, −1.25 pp

6.4. Results

Table 8 compares the steady-state effects across adjustment mechanisms, revealing stark differences in macroeconomic and welfare outcomes.

Raising VAT to offset lost revenue reduces output by 0.46% and consumption by 1.19%, generating a welfare loss (CEV) of −1.19%. Increasing employer PRSI has the largest negative effects, reducing output by 4.40% and consumption by 4.29%. In contrast, reducing government consumption produces only a small output decline (−0.77%) but

Table 8: *Scenario Comparison: Mnc Withdrawal*

Scenario	ΔY (%)	ΔC (%)	CEV (%)
mnc_shock_vat	-0.46%	-1.19%	-1.19%
mnc_shock_payroll	-4.40%	-4.29%	-4.29%
mnc_shock_spending	-0.77%	+4.16%	+4.16%
mnc_shock_transfers	+0.00%	-0.00%	-0.00%
mnc_shock_mixed	-0.42%	+0.98%	+0.98%

Note: All values relative to baseline. CEV = Consumption Equivalent Variation.

a consumption *gain* of 4.16%, as private consumption substitutes for public spending. Reducing transfers has negligible aggregate effects but would have significant distributional consequences. A mixed adjustment combining VAT increases, spending cuts, and transfer cuts produces moderate effects with a small welfare gain (+0.98%). Table 9 presents detailed macroeconomic effects for the VAT adjustment scenario.

Table 9: *Macroeconomic Effects: Mnc Shock Vat*

Variable	Baseline	Reform	% Change
Output (Y)	0.6307	0.6278	-0.46%
Consumption (C)	0.3861	0.3815	-1.19%
Capital (K)	2.2584	2.2852	+1.19%
Labor (L)	0.3173	0.3131	-1.33%
Interest Rate (r)	0.0452	0.0437	-3.39%
Wage (w)	1.2919	1.3033	+0.89%

6.5. Discussion

The results highlight several important findings for Irish fiscal policy. The choice of fiscal adjustment mechanism has first-order effects on welfare: payroll tax increases are particularly costly (-4.3% CEV), while spending cuts can actually improve welfare ($+4.2\%$ CEV) by shifting resources from public to private consumption. Payroll tax increases generate the largest output losses because they directly distort the labor-leisure margin, and Ireland's already-high marginal tax rates on labor income amplify these effects.

The VAT adjustment produces modest welfare losses (-1.2% CEV) despite the large rate increase required, reflecting the relatively low distortionary cost of consumption taxes compared to labor taxes. The mixed adjustment scenario produces a small welfare gain ($+1.0\%$ CEV) by combining the efficiency of spending cuts with smaller increases in distortionary taxes. These findings underscore the importance of fiscal preparedness—a sudden MNC revenue shock would require difficult choices, and the adjustment path matters significantly for long-run welfare.

7. Comparison of Policy Effects

Table 10 summarizes the steady-state effects of the policy experiments for which we have completed simulations.

Table 10: *Summary of Policy Effects (Steady State, % change from baseline)*

Variable	PRSI Cut (-3pp)	USC Abolition (full removal)	MNC Shock (VAT adjustment)
Output (Y)	+1.95%	+2.63%	−0.46%
Consumption (C)	+1.89%	+2.27%	−1.19%
Capital (K)	+2.07%	+3.59%	−0.59%
Labor (L)	+1.89%	+2.12%	−0.40%
Interest rate (r)	−0.24%	−1.52%	+0.38%
Wage (w)	+0.06%	+1.14%	−0.06%
CEV	+1.89%	+2.27%	−1.19%

Note: PRSI cut is a 3pp reduction in employer contribution. USC abolition uses EUROMOD-estimated age- and income-varying tax functions. MNC Shock assumes 3.75% of GDP revenue loss offset by VAT increase.

Employer PRSI reduction produces substantial positive effects, with approximately 2% gains in output, consumption, and labor supply; the welfare gain (CEV of +1.89%) reflects reduced labor tax distortions. USC abolition generates the largest positive effects, with 2.63% gains in output and 2.27% gains in consumption. The welfare gain (CEV of +2.27%) reflects reduced distortions from both effective and marginal tax rate reductions, with capital accumulation (+3.59%) particularly strong due to increased disposable income and saving.

Both labor tax reforms dominate the MNC shock scenario. While the MNC revenue shock with VAT adjustment produces modest negative effects (−1.19% CEV), reductions in labor taxation generate substantial welfare gains, highlighting the importance of maintaining low labor tax wedges for economic efficiency. Capital accumulation responses differ across policies: USC abolition generates the largest capital stock increase (+3.59%), reflecting both reduced effective tax rates and reduced marginal rates on capital income, while PRSI cuts have more modest capital effects (+2.07%) as they primarily affect labor taxation. Among MNC shock scenarios, the choice of fiscal adjustment mechanism matters significantly—spending cuts produce better welfare outcomes than tax increases, though with different distributional implications (see Table 8).

8. Conclusion

This paper has presented DynÉire, a dynamic overlapping generations model calibrated to the Irish economy. The model captures key features of Ireland's fiscal environment, including its vulnerability to MNC tax revenue shocks and the structure of labor taxation including the USC and PRSI.

We find that a 3 percentage point cut in employer PRSI generates substantial welfare gains (+1.9% CEV), with output increasing 1.95% through both increased labor supply (+1.89%) and capital accumulation (+2.07%). The reduction in the labor tax wedge stimulates employment and investment.

Complete removal of the Universal Social Charge generates the largest welfare gains (+2.27% CEV) among the policies examined. Using EUROMOD-estimated tax functions that vary by age and income, we find output increases by 2.63%, driven by labor supply growth (+2.12%) and particularly strong capital accumulation (+3.59%). The fiscal cost is approximately 9% of baseline tax revenue, partially offset by the expanded tax base.

A sudden loss of MNC corporate tax revenue (3.75% of GDP) would require significant fiscal adjustment, with the choice of instrument having first-order welfare effects. Spending cuts produce better aggregate welfare outcomes than tax increases (+4.2% vs. −4.3% CEV for payroll taxes), though with different distributional implications.

Future work will extend the analysis in several directions:

- **Full Transition Path Dynamics**, see below.
- **Open economy extensions**: Explicit modeling of international capital flows, foreign direct investment, and the distinction between domestic and foreign-owned capital, particularly relevant for Ireland's MNC-intensive economy.
- **Additional policy scenarios**: Analysis of other tax and benefit reforms, including income tax rate changes, property tax reforms, and pension system adjustments.
- **Sensitivity and robustness**: Comprehensive sensitivity analysis across key behavioral parameters (Frisch elasticity, intertemporal elasticity of substitution) and alternative calibrations.
- **Extended EUROMOD integration**: Applying the EUROMOD-OG integration methodology (demonstrated for USC) to other policy experiments, including the MNC shock

and PRSI scenarios, to capture distributional effects across the full income distribution.

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A. Data Sources

Table 11: *Primary Data Sources for Calibration*

Data	Source	Series/Table
Population by age	CSO	PEA11
Fertility rates	CSO	VSA02
Mortality rates	Human Mortality Database	Ireland tables
Migration	CSO	PEA15
National accounts	CSO	N1903
Government finance	CSO	GFQ02
Government debt	Eurostat	gov_10dd_edpt1
Earnings	CSO	EHA01, EHA11
Income distribution	Eurostat/CSO	EU-SILC
Tax statistics	Revenue Commissioners	Annual reports
Social welfare	DSP	Statistical reports
Wealth distribution	ECB	HFCS Ireland

B. Model Solution Algorithm

The model is solved using the following algorithm:

1. **Steady State**:
   (a) Guess initial values for (r, w)
   (b) Solve household problems by backward induction
   (c) Aggregate individual decisions
   (d) Check market clearing conditions
   (e) Update guesses and iterate until convergence
2. **Transition Path**:
   (a) Compute initial and terminal (new) steady states
   (b) Guess price path $\{r_t, w_t\}_{t=1}^{T}$

(c) Solve household problems along transition

(d) Aggregate and check market clearing

(e) Update price path and iterate

The computational implementation uses the OG-Core package, which provides efficient numerical methods for these calculations including parallel processing via Dask.

C. Sensitivity Analysis

[TO BE COMPLETED]

Key parameters for sensitivity analysis:

- Frisch elasticity: [0.2, 0.4, 0.6]
- Intertemporal elasticity of substitution: [0.3, 0.5, 0.7]
- Profit-shifting elasticity: [0.5, 1.0, 2.0, 3.0]
- MNC share of corporate tax: [0.65, 0.75, 0.85]

D. Transition Path Dynamics

[Under Construction]

The results presented in this paper focus on steady-state comparisons—that is, comparisons between the initial long-run equilibrium and the new long-run equilibrium following a policy change. While steady-state analysis reveals the ultimate effects of policy reforms, it does not capture the adjustment dynamics: how the economy transitions from one equilibrium to another over time.

Transition Path Iteration (TPI) is a computational method for solving the full dynamic path of the economy following a policy change. The algorithm works as follows:

1. Compute the initial steady state (pre-reform) and terminal steady state (post-reform)
2. Guess an initial path for prices $\{r_t, w_t\}_{t=1}^{T}$ connecting the two steady states
3. Given these prices, solve each household's optimization problem over the transition
4. Aggregate household decisions to obtain implied factor supplies
5. Check market clearing conditions and update the price path

6. Iterate until convergence

Transition dynamics are valuable for several reasons:

- **Short-run vs. long-run effects**: Some policies may have different effects in the short run than in the long run. For example, capital accumulation takes time, so the full output effects of tax reforms may not materialize for decades.
- **Welfare across generations**: Different cohorts experience policy changes differently depending on when in their lifecycle the reform occurs. TPI allows computation of cohort-specific welfare effects.
- **Fiscal sustainability**: Transition paths reveal the time profile of government revenues and expenditures, which is crucial for assessing fiscal sustainability.
- **Political economy**: Understanding the short-run costs and long-run benefits of reforms is essential for policy design and political feasibility.

We are currently working on implementing TPI for the DynÉire model. Once complete, this will enable analysis of how quickly the Irish economy adjusts to policy changes and how the costs and benefits of reform are distributed across time and generations.